S. Saketha Ramanujam **Samavedam**

s3r1@duck.com • linkedin.com/in/sakethramanujam

Education

Master of Science in Electrical Engineering **A.S.U., Tempe, AZ**
Thesis: Ray Casting based Correction Algorithm for Terahertz Non-Line-of-Sight Imaging *CGPA: 3.43/4.0*

Bachelor of Technology in Electronics and Communications Engineering **G.V.P.C.E (A), Vizag, IN**
Thesis: Application of Random Forest Algorithm for Classification of Photonic Sensor Data *CGPA: 7.92/10*

Relevant Graduate Coursework

Microwave Circuit Design ▪ Microwave Solid-State Circuit Design ▪ Antenna Analysis and Design ▪ Millimeter Wave & THz Measurements ▪ Principles of Signal & Power Integrity ▪ Coherent Optics ▪ Random Signal Theory

Technical Skills

Languages: C++ ▪ C ▪ Python ▪ Go ▪ Bash ▪ MATLAB ▪ JavaScript ▪ SQL
Image Processing & Vision: ASIL Functional Safety ▪ OTP/EEPROM Provisioning ▪ ISP Pipelines ▪ Raw Image Processing ▪ MTF/PTC/Noise Characterization ▪ OpenCV
Embedded & Sensors: Automotive CMOS Image Sensors ▪ NVIDIA Jetson/Orin ▪ Camera Framegrabbers ▪ v4l2 Drivers ▪ MIPI CSI-2 ▪ I2C ▪ SPI ▪ UART ▪ Arduino ▪ Raspberry Pi ▪ SDR
Signal Processing & ML: Quadrature Signal Decomposition ▪ Statistical Feature Extraction ▪ Random Forest Classification/Regression ▪ scikit-learn
Infrastructure: Docker ▪ Nginx ▪ Kubernetes ▪ Kafka ▪ CI/CD (GitHub Actions) ▪ Terraform ▪ GCP
Test & Measurement: VNA ▪ Spectrum Analyzer ▪ Tescom DAB Tester ▪ Vector Signal Generators ▪ Trioptics MTF Studio
EM & RF: DAB ▪ FM ▪ FM/HD ▪ SXM Satellite Radio ▪ Ansys HFSS ▪ Keysight ADS ▪ LabView

Professional Experience

Camera Systems Engineer | Tesla Motors Inc. Palo Alto, CA Jun 2025 – Present
Awards: Recipient of Top 20% Merit award for the year 2025 under Camera Systems Group
Technologies: C++, C, Go, Python, Bash, Nginx, Docker, Automotive CMOS Image Sensors, Camera Framegrabbers, NVIDIA Jetson/Orin

- Co-authored **Image Sensor Functional Safety**: built fault injection infrastructure for ASIL-rated sensors across two generations and the sensor-to-cloud telemetry pipeline (OTP, calibration angles, pixel shift, safety registers) for fleet-wide monitoring across **millions of vehicles**.
- Built a **camera capture API server** (FastAPI) layer over custom built C MTF capture client bridging Tesla's camera libraries with Trioptics MTF Studio — remote capture, exposure control, 12-bit RAW unpacking for end-of-line testing. Added dark noise mode to the frame-grabber GRPC client for band noise quantification.
- Led cross-platform **camera extrinsic validation** across all Tesla vehicles (Robotaxi, Cybertruck, Model 3/Y, Model S/X), verifying orientation and calibration against mechanical drawings for FSD perception geometry.
- Currently leading a cross-functional team (mechanical, electrical, systems) developing an **off-gassing detection fixture** for production lines and service centers. Wrote haze/streak detection routines using image processing algorithms for factory camera checks and vehicle upkeep mode. This instrument **cuts down precision inspection time by $>$ 80%.**
- Built **imager UUID authentication** against cloud EEPROM records over I2C, preventing unverified sensors from operating in safety-critical autonomous driving systems.
- Architected Tesla's **Gen 2 image quality testing platform**: distributed multi-tenant system with edge agents, priority scheduling, lab instrument discovery, and calibration tracking. Serves the Camera Hardware organization.
- Built a **multi-camera data collection platform** for Tesla's Optimus robot: C/C++ v4l2 capture over MIPI CSI-2, CUDA Bayer-to-YUV encoding, custom sensor kernel drivers, Flask web interface for synchronized 4-camera capture on NVIDIA Jetson.

Radio Systems Engineer | Tesla Motors Inc. Palo Alto, CA Dec 2022 – Jun 2025

Awards: Recipient of Top 20% Merit award for the year 2024 under Electronic Systems Group

Technologies: C++, Python, Bash, Docker, Kubernetes, Kafka, Tescom DAB Tester, Vector Signal Generators

- Deployed fleet-wide signed scripts disabling Traffic Announcements on EU vehicles, resolving a customer-facing issue across the European fleet.
- **Owned Gen3 radio tuner validation** across 9 firmware epics (Gen2/Gen3 bring-up through V8–V11) for Harman and Hirschmann suppliers, from bring-up through production across **4 Tesla factories** (Fremont, Austin, Berlin, Shanghai).
- Led **global market radio certification** across 12+ countries — multi-day European field tests validating DAB/FM reception, station discovery, ensemble handoff, and blending thresholds.
- Executed 1000-cycle OTA tests, low-power mode characterization (V7–V10), antenna diagnostics, and thermal analysis. Built automated daily failure tracking by country config, VIN, and firmware version.
- Designed the **unified radio tuner connection test** across all platforms and factories. Shipped the **Radio Tuner Debug UI** in vehicle firmware for real-time tuner diagnostics in developer settings.
- Created 17 internal tools (CLI, field test web UI, REPL, data scrapers, Chrome extension). Authored the team's on-boarding guide and knowledge base across all tuner firmware versions.
- Built the **factory audio data pipeline**: Kafka consumers for end-of-line test data from 3 factories with S3 archival, migrated on-premise to AWS. Shipped the factory audio sweep test in vehicle firmware.

Teaching Assistant, EEE341, EEE445 | A.S.U Tempe, AZ Aug 2022 – Dec 2022

Technologies: Engineering Electromagnetics, Microwave Engineering

- TA for Microwave Circuit Design and Engineering Electromagnetics. Led office hours and lab sessions on VNA measurements and microstrip circuit characterization.

Electronic Systems Intern | Tesla Motors Inc. Palo Alto, CA Jan 2022 – Aug 2022

Technologies: Python, Bash, Docker, Kubernetes, Kafka, AWS S3

- Architected Kafka data pipelines for factory audio end-of-line data from Fremont and Austin with S3 archival and engineering file server.
- Validated Gen2/Gen3 radio tuners across S, 3, X, Y platforms for R12.4 firmware — OTA, sleep tests, worldwide coordinated tuner tests.
- Implemented speaker-dependent frequency limiting (ADSP BSR) in vehicle firmware.

Student Researcher | Terahertz Research Lab, A.S.U. Summer 2021 – April 2024

Technologies: Radar Imaging, OpenCV, FMCW Radar, Beam-forming, MATLAB, Python

- Developed a **ray casting-based correction algorithm** for THz non-line-of-sight imaging: classified line-of-sight vs. non-line-of-sight objects using angular ray casting, detected reflective surfaces via Canny edge detection and probabilistic Hough line transforms, and corrected object placement through mirror folding across multiple bounce generations.
- Built radar image formation pipelines applying **inverse Fourier transforms**, angle-of-arrival, and time-of-flight processing to FMCW radar data from TI AWR2243 cascaded radar. Implemented in MATLAB and Python (NumPy, SciPy, OpenCV, scikit-image).

Coding & Research Aide | ASU Biodesign Institute (Dr. Leland Hartwell) Jun 2021 – Jan 2022

Technologies: JavaScript, WebGazer.js, Node.js, Express, Computer Vision, PsychoPy/Pavlovia

- Built **Postimate**, a webcam-based gaze tracking and heatmap generation system for Nobel laureate Dr. Leland Hartwell's psychology research at ASU Biodesign Institute. Designed the camera calibration pipeline (9-point grid), real-time eye tracking via computer vision, and gaze-to-screen coordinate mapping for experiment stimulus analysis.
- Integrated with **PsychoPy/Pavlovia** experiment platform for behavioral study data collection, enabling researchers to measure visual attention patterns through imaging sensor input.

Graduate Services Assistant | Calhoun Lidar Lab, A.S.U. Mar 2021 – Jan 2022

Technologies: MEMS LiDAR, SPAD, Time-to-Digital Converters, I2C, Arduino, Python

- Designed subsystems for MEMS-based navigation LiDAR using Single-Photon Avalanche Diodes and Time-to-Digital Converters. Implemented sensor communication over I2C and serial interfaces. Managed an inter-disciplinary team of 3.

Data Acquisition Assistant | **A.S.U.** Feb 2021 – Mar 2021

Technologies: Python, Scrapy, Web Scraping

- Built parameterized web scraping tools for data acquisition from public databases, including the TRAC database on ICE removals.

R&D Engineer | **Scientific and Industrial Research Centre, GVP** Dec 2019 – Dec 2020

Technologies: Python, C, Raspberry Pi, TCP/IP Sockets, scikit-learn, LabView, MATLAB, Electron

- Built the **AUM photonic sensor software ecosystem**: real-time data acquisition dashboard (Electron + Python), signal processing library, experiment control web applications (Flask), and ML prediction pipeline for atmospheric gas classification from optical backscatter sensor data.
- Developed **quadrature signal decomposition** (magnitude and direction from in-phase and quadrature photodetector channels), backward differencing, and windowed statistical feature extraction (35 features across 5 channels) for MultiOutput RandomForest models predicting 19 atmospheric parameters including CO, NH3, O3, NO2, SO2, PM2.5, and PM10.
- Conducted **field calibration campaigns** in Nasik and Visakhapatnam, co-locating AUM sensors with CAAQMS reference stations and Kestrel weather instruments for ground-truth model training and validation.
- Designed structural health analysis experiments for metallic structures using opto-electronic devices.

Google Summer of Code Mentor | **BeagleBoard.org** May 2020 – Aug 2020

Technologies: C, Linux, PocketBeagle, BeagleBone Black, TI AM335x PRU, RemoteProc

- Co-mentored the **PRU Bi-directional Bus** project: implementation of a bidirectional communication bus using the TI AM335x Programmable Real-time Unit (PRU) subsystem on PocketBeagle for deterministic, low-latency I/O expansion independent of the ARM Linux core.

Software Engineering Intern | **Rorodata, Bengaluru, IN** Jun 2019 – Dec 2019

Technologies: Python, Docker, Nginx, Terraform, GCP, GitHub Actions, Apache Airflow

- Built automated test suites (Pytest, Selenium, Nginx) and CI/CD pipelines using Google Cloud Build, GitHub Actions, and Terraform for a data science deployment platform.
- Evaluated Apache Airflow for ML workload orchestration. Performance-tested APIs on GCP (App Engine, Compute Engine, Cloud Run) using Apache Bench and Locust.
- Deployed internal Python package server (devpi) and built web scrapers and CLI tools for data collection.

Summer Research Intern | **Scientific and Industrial Research Centre, GVP** Apr 2018 – Jun 2018

Technologies: Python, C, Raspberry Pi, Arduino, Web Sockets, Firebase, LabView, MATLAB

- Developed wireless sensor data transmission client using TCP/IP socket programming on Raspberry Pi for real-time photonic sensor data streaming.
- Built custom image upload and analysis engine (Google Firebase) for microbial monitoring in water samples using laser-water interaction imaging.
- Designed a trans-impedance amplifier circuit for the AUM photonic system and a band-pass filter for noise cancellation in opto-electronically recorded audio signals.

Academic Projects

Non Line of Sight Imaging with mmWave Cascaded Radar Fall 2021 – Summer 2024

Automated calibration and real-time imaging with TI AWR2243 cascaded radar. Developed image processing algorithms for non-line-of-sight scene reconstruction using electromagnetic wave propagation and specular reflection modeling.

2-Axis Scanned LiDAR with Arduino Summer 2021

Built a servo-controlled LiDAR scanner using Garmin LIDAR-Lite over I2C and Arduino. Spherical-to-Cartesian coordinate transforms with real-time 3D point cloud visualization.

Transverse Deflection Structure in HFSS Spring 2021

Designed and simulated an RF cavity resonator centered at 9 GHz in Ansys HFSS for study of TEM wave propagation in linear accelerators. Eigenmode and driven-mode solver analysis.

Webcam Gaze Tracker with OpenCV 2020

Real-time eye gaze tracking application using OpenCV Haar cascades and Hough circle detection for pupil localization. Maps camera sensor coordinates to screen space for heatmap visualization of visual attention.

Vision Aided Left Behind Object Detection Spring 2019

Trained a YOLOv2-based object detector for abandoned baggage detection in crowded public areas using a static

camera and Linux-based prototype.

Photonic Sensor Air Quality Monitoring (AUM) 2018 – 2020

End-to-end photonic sensing platform: Raspberry Pi data acquisition over TCP sockets, quadrature signal processing, statistical feature extraction, and ensemble ML prediction for 19 atmospheric parameters. Field-calibrated across two cities in India.

Achievements

- Two-time recipient of Tesla's **Top 20% Merit Award** (2024 Electronic Systems, 2025 Camera Systems).
- Awarded Masters Opportunity for Research in Engineering (ASU MORE) Grant for Fall 2021.
- Speaker, DevConf.IN 2019 (Red Hat): "The Subtle Art of Backward Differencing" — finite difference methods for sensor signal feature engineering.

Volunteering and Outreach

- **GitHub Campus Expert** — organized Hacktoberfest Vizag (2018), MLH Local Hack Day at GVP (2018). 2018 – Present
- **Google Summer of Code Mentor**, BeagleBoard.org — PRU Bi-directional Bus on TI AM335x 2020
- **Founder, GVP AI Club** — created curriculum and taught workshops on Python for Data Science, Machine Learning, and Deep Learning to 20+ students. Authored organizational governance and contribution guidelines. 2018 – 2019
- National Service Scheme Volunteer, Gayatri Vidya Parishad 2016 – 2018

Publications

- Ray Casting based Correction Algorithm for Terahertz Non-Line-of-Sight Imaging